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Inventor(s)

TUOMIRANTA; Arttu et al.

METHOD FOR DETERMINING AN ORIENTATION OF A PHOTOVOLTAIC PANEL TO REDUCE THE DAMAGE CAUSED BY A HAIL EVENT

Abstract

The invention relates to a method for determining an orientation of a photovoltaic panel to reduce the damage caused by a hail event, the method being computer-implemented and comprising the following phases receiving input data relative to characteristics of a hail event, the input data comprising at least the mass of a hailstone and a hailstone velocity vector, and determining an orientation of the photovoltaic panel as a function of the input data, in order to protect elements underneath the photovoltaic panel while preventing damages to occur on the photovoltaic panel, the determined orientation corresponding to an intermediate orientation of the photovoltaic panel between a plurality of possible intermediate orientations.

Inventors: TUOMIRANTA; Arttu (PARIS LA DEFENSE CEDEX, FR), CHAPON; Julien (PARIS LA DEFENSE CEDEX, FR), DRAHI; Etienne (PARIS LA DEFENSE CEDEX, FR), CUIF-SJOSTRAND; Marianne (PARIS LA DEFENSE CEDEX, FR)

Applicant: TotalEnergies OneTech (COURBEVOIE, FR)

Family ID: 90368674

Assignee: TotalEnergies OneTech (COURBEVOIE, FR)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority of European Patent Application No. 24305363.4 filed Mar. 11, 2024. The entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The present invention concerns a method for determining an orientation of a

[0003] photovoltaic panel to reduce the damage caused by a hail event. The present invention also concerns an associated computer program product. The present invention also relates to an associated readable information carrier.

BACKGROUND OF THE INVENTION

[0004] The production of electricity from renewable energy sources is essential for our societies. To this end, dedicated installations have been developed, in particular photovoltaic installations, which allow electricity to be produced from solar energy. Photovoltaic installations typically comprise several photovoltaic panels which are deployed over the surface of a given area.

[0005] Fields are areas of interest for the installation of photovoltaic panels because of the large area of such fields. Hence, the photovoltaic panels can coexist with crops of a field.

[0006] In such areas, the photovoltaic panels, as well as the crops, are subject to hail events, which can severely damage them.

[0007] The crops are sometimes disposed under the photovoltaic panels so as to protect them from a hail event.

[0008] Nevertheless, depending on the orientation of the hail and the position of the panels when those ones are not fixed, the crops can still be damaged. In addition, this does not prevent the photovoltaic panels from being damaged.

SUMMARY

[0009] There exists a need for a means enabling to reduce the damages caused by a hail event on both photovoltaic panels and elements underneath the photovoltaic panel.

[0010] To this end, the invention relates to a method for determining an orientation of a photovoltaic panel to reduce the damage caused by a hail event, the method being computer-implemented and comprising the following phases: [0011] receiving input data relative to characteristics of a hail event, the input data comprising at least the mass of a hailstone and a hailstone velocity vector, and [0012] determining an orientation of the photovoltaic panel as a function of the input data, in order to protect elements underneath the photovoltaic panel while preventing damages to occur on the photovoltaic panel, the determined orientation corresponding to an intermediate orientation of the photovoltaic panel between a plurality of possible intermediate orientations, each possible intermediate orientation being such that the angle, called hail impact angle, between the normal to the surface of the photovoltaic panel and the hailstone velocity vector, is strictly comprised between 0 degrees and 90 degrees modulo π , the orientation of the photovoltaic panel corresponding to a hail impact angle of 0 degrees enabling to maximize the protection of elements underneath the photovoltaic panel without preventing damages to occur on the photovoltaic panel, the orientation of the photovoltaic panel corresponding to a hail impact angle of 90 degrees enabling to maximize the prevention of damages on the photovoltaic panel without protecting elements underneath the photovoltaic panel.

[0013] The method according to the invention may comprise one or more of the following features considered alone or in any combination that is technically possible: [0014] the method comprises a phase of sending a command to an actuator of the photovoltaic panel enabling to adapt the orientation of the photovoltaic panel in conformity with the determined orientation; [0015] the determination phase comprises the determination of the orientation of the photovoltaic panel so as to fulfill a kinetic energy criterion, the kinetic energy criterion stating that the kinetic energy received by the photovoltaic panel from a hailstone is below a maximum predetermined kinetic energy, enabling to prevent damages to occur on the photovoltaic panel; [0016] the determination phase comprises the steps of: [0017] calculating the hailstone maximum velocity corresponding to the maximum predetermined kinetic energy, [0018] calculating the hailstone velocity as a function of the input data, [0019] comparing the hailstone velocity to the hailstone maximum velocity, and [0020] determining the orientation of the photovoltaic panel as a function of the input data and of the result of the comparison; [0021] when the hailstone velocity is above the hailstone maximum velocity, the step of determining the orientation of the photovoltaic panel comprises calculating the minimum allowable hailstone

impact angle on the basis of the hailstone maximum velocity and the hailstone velocity, and calculating the orientation of the photovoltaic panel as a function of the minimum allowable hailstone impact angle and of the input data; [0022] the orientation of the photovoltaic panel is quantified by a tilt angle for the photovoltaic panel and an azimuth angle for the photovoltaic panel enabling to perform dual axis tracking, the step of determining the orientation of the photovoltaic panel comprising fixing one of the two angles and determining the other angle depending on the input data and of the result of the comparison; [0023] the input data also comprise the azimuth of the wind, the step of determining the orientation of the photovoltaic panel comprising setting the tilt angle of the photovoltaic panel at a given value, and [0024] when the hailstone velocity is above the hailstone maximum velocity, determining two solutions for the azimuth angle as a function of the following equation:

$$[00001] \quad \gamma_{\text{sub.poa}} = \gamma_{\text{sub.w}} \pm \left(\frac{\cos(\theta_{\text{sub.hail,poa,min}}) - \cos(\beta_{\text{sub.poa}}) \cos(\theta_{\text{sub.hail}})}{\sin(\gamma_{\text{sub.poa}}) \sin(\gamma_{\text{sub.w}})} \right)$$

[0025] where: [0026] $\gamma_{\text{sub.poa}}$ is the azimuth angle of the photovoltaic panel, [0027] $\gamma_{\text{sub.w}}$ is the azimuth of the wind, [0028] $\theta_{\text{sub.hail,poa,min}}$ is the minimum allowable hailstone impact angle, [0029] $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector and a vertical axis, and [0030] $\beta_{\text{sub.poa}}$ is the tilt angle of the photovoltaic panel, [0031] when the hailstone velocity is below the hailstone maximum velocity, setting the azimuth angle of the photovoltaic panel as being equal to the azimuth of the wind; [0032] the tilt angle of the photovoltaic panel is set as being equal to the hailstone velocity angle, the hailstone velocity angle being the angle between the hailstone velocity vector and a vertical axis; [0033] the input data also comprise the azimuth of the wind, the step of determining the orientation of the photovoltaic panel comprising setting the azimuth angle of the photovoltaic panel at a given value, and [0034] when the hailstone velocity is above the hailstone maximum velocity, determining two solutions for the tilt angle as a function of the following equation:

$$[00002] \cos(\theta_{\text{sub.hail,poa,min}}) = \cos(\beta_{\text{sub.poa}}) \cos(\theta_{\text{sub.hail}}) + \sin(\beta_{\text{sub.poa}}) \sin(\theta_{\text{sub.hail}}) \cos(\gamma_{\text{sub.w}} - \gamma_{\text{sub.poa}})$$

[0035] where: [0036] $\gamma_{\text{sub.poa}}$ is the azimuth angle of the photovoltaic panel, [0037] $\gamma_{\text{sub.w}}$ is the azimuth of the wind, [0038] $\theta_{\text{sub.hail,poa,min}}$ is the minimum allowable hailstone impact angle, [0039] $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector and a vertical axis, and [0040] $\beta_{\text{sub.poa}}$ is the tilt angle of the photovoltaic panel, [0041] when the hailstone velocity is below the hailstone maximum velocity, setting the tilt angle of the photovoltaic panel as being equal to the hailstone velocity angle, the hailstone velocity angle being the angle between the hailstone velocity vector and a vertical axis; [0042] the azimuth angle of the photovoltaic panel is set as being equal to the azimuth of the wind; [0043] the orientation of the photovoltaic panel is quantified by an angle of rotation of a single axis oriented at a tilt angle and at an azimuth angle, enabling to perform single axis tracking, the step of determining the orientation of the photovoltaic panel comprising: [0044] when the hailstone velocity is below the hailstone maximum velocity, the hail impact angle, between the normal, to the surface of the photovoltaic panel and the hailstone velocity vector is calculated the following way:

[00003]

$$\theta_{\text{sub.hail,poa}} = \cos^{-1}(\cos(\omega_{\text{sub.prot}}) \cos(\gamma_{\text{sub.axis}}) \cos(\theta_{\text{sub.hail}}) + \sin(\omega_{\text{sub.prot}}) \sin(\theta_{\text{sub.hail}}) \cos(\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}})) + \sin(\omega_{\text{sub.prot}}) \sin(\theta_{\text{sub.hail}}) \sin(\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}}),$$

$$\text{where } \gamma_{\text{sub.prot}} = \tan^{-1}(X) + \gamma_{\text{sub.axis}} = \tan^{-1}\left(\frac{\sin(\theta_{\text{sub.hail}}) \cos(\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}})}{\cos(\gamma_{\text{sub.axis}}) \cos(\theta_{\text{sub.hail}}) + \sin(\gamma_{\text{sub.axis}}) \sin(\theta_{\text{sub.hail}}) \cos(\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}})}\right) + \gamma_{\text{sub.axis}}, \text{ where}$$

$$0^\circ, \text{ if } X = 0, \text{ or if } X > 0 \text{ and } (\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}}) > 0,$$

$$\text{or if } X < 0 \text{ and } (\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}}) < 0$$

= {

$$+180^\circ, \text{ if } X < 0 \text{ and } (\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}}) > 0$$

$$-180^\circ, \text{ if } X > 0 \text{ and } (\gamma_{\text{sub.w}} - \gamma_{\text{sub.axis}}) < 0$$

[0045] where: [0046] $\omega_{\text{sub.prot}}$ is the protection angle of rotation of the single axis, [0047] $\beta_{\text{sub.axis}}$ is the tilt angle of the single axis, [0048] $\gamma_{\text{sub.axis}}$ is the azimuth angle of the single axis, [0049] $\gamma_{\text{sub.w}}$ is the azimuth of the wind, [0050] $\theta_{\text{sub.hail,poa,min}}$ is the minimum allowable hailstone impact angle, and [0051] $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector and a vertical axis, [0052] when the hailstone velocity is below the hailstone maximum velocity, and consequently, when the hail impact angle, between the normal, to the surface of the photovoltaic panel and the hailstone velocity vector, is always greater than or equal to the minimum allowable hailstone impact angle, the rotation angle of the single axis is set as being equal to the value; [0053] when the hail impact angle, between the normal to the surface of the photovoltaic panel, which is rotated at $\omega_{\text{sub.prot}}$, and the hailstone velocity vector, is below the minimum allowable

hailstone impact angle, two solutions are determined for the rotation angle ω of the single axis as a function of the following equation.

[00004]

$$\sin_{\text{hail, poa, min}} = \cos^{-1}(\cos(\cos_{\text{axis}} \cos_{\text{hail}} + \sin_{\text{axis}} \sin_{\text{hail}} \cos(\omega - \text{axis})) + \sin_{\text{axis}} \sin_{\text{hail}} \sin(\omega - \text{axis}))$$

[0054] where: [0055] ω is the angle of rotation of the single axis. [0056] the hailstone velocity vector comprises a vertical velocity component and a horizontal velocity component, the horizontal velocity component being approximated as being the velocity of the wind; [0057] the vertical velocity component is obtained on the basis of the following formula:

[00005]
$$u_z = \sqrt{\frac{8 \times \rho_{\text{hail}} \times R_{\text{hail}} \times g}{3 \times \rho_{\text{air}} \times 0.44}}$$

[0058] Where: [0059] $\rho_{\text{sub.hail}}$ is the density of the hail, [0060] $R_{\text{sub.hail}}$ is the radius of the hail, [0061] g is the acceleration of gravity, and [0062] $\rho_{\text{sub.air}}$ is the density of air.

[0063] The invention also relates to a computer program product comprising a readable information carrier having stored thereon a computer program comprising program instructions, the computer program being loadable onto a data processing unit and causing a method as previously described to be carried out when the computer program is carried out on the data processing unit.

[0064] The invention also relates to a readable information carrier on which is stored a computer program product as previously described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0065] The invention will be better understood, based on the following description, given solely as an example and made in reference to the appended drawings in which:

[0066] FIG. 1 is a schematic view of an example of a photovoltaic panel under which there are crops, the orientation of the photovoltaic panel being controlled by an actioner as a function of a command received from a calculator,

[0067] FIG. 2 is a flowchart of an example of implementation of a method for determining an orientation of a photovoltaic panel to reduce the damage caused by a hail event,

[0068] FIG. 3 is a schematic representation of a photovoltaic panel and of a hailstone arriving on the photovoltaic panel, such representation illustrating the vectors and angles considered in some calculations, and [0069] FIG. 4 is a schematic representation of a photovoltaic panel in a one-axis tracking geometry.

DETAILED DESCRIPTION

[0070] A photovoltaic panel **10** protecting elements **12**, which are under the photovoltaic panel **10**, is illustrated on FIG. 1. On this FIG. 1, the orientation of the photovoltaic panel **10** is controlled by an actioner **14** receiving a command from a calculator **16**.

[0071] The elements **12** are for example crops, such as vegetable, fruits, trees or flowers. More generally, the elements **12** can also be cars or anything of value that may be damaged by hail.

[0072] The photovoltaic panel **10** comprises a photovoltaic surface **10A** and a mounting structure **10B**. The photovoltaic surface **10A** defines a plane, called POA in what follows. The photovoltaic surface **10A** is a surface at the front or at the rear of the photovoltaic panel **10**. The photovoltaic surface **10A** is typically composed of several photovoltaic modules, each photovoltaic module comprising several photovoltaic cells.

[0073] The mounting structure **10B** is suitable to be moved by the actioner **14** so as to move the photovoltaic surface **10A**.

[0074] The actioner **14** is suitable for two-axis tracking or for one-axis tracking depending on the nature of the photovoltaic panel **10**.

[0075] A one-axis tracking (or single-axis tracking) implies a move on only one axis of movement, while two-axis tracking (or dual-axis tracking) implies a move on two axes. Solar panels with a single-axis tracker usually only move from East to West, whereas solar panels with a dual-axis tracker usually move from East to West and from North to South.

[0076] The calculator **16** is preferably a computer.

[0077] More generally, the controller **16** is a computer or computing system, or similar electronic computing device adapted to manipulate and/or transform data represented as physical, such as electronic, quantities within the computing system's registers and/or memories into other data similarly represented as physical quantities within the computing system's memories, registers or other such information storage, transmission

or display devices.

[0078] For example, the calculator **16** comprises a processor comprising a data processing unit, memories and a reader for information media, and eventually, a human machine interface, such as a keyboard, and a display.

[0079] In an example of implementation, the calculator **16** interacts with a computer program product. The computer program product comprises an information medium.

[0080] The information medium is a medium readable by the calculator **16**, usually by the data processing unit. The readable information medium is a medium suitable for storing electronic instructions and capable of being coupled to a computer system bus.

[0081] By way of example, the information medium is a USB key, a floppy disk or flexible disk (of the English name “Floppy disc”), an optical disk, a CD-ROM, a magneto-optical disk, a ROM memory, a memory RAM, EPROM memory, EEPROM memory, magnetic card or optical card.

[0082] On the information medium is stored the computer program comprising program instructions.

[0083] The computer program is loadable on the data processing unit and is adapted to entail the implementation of a method for determining an orientation of the photovoltaic panel **10** to reduce the damage caused by a hail event, when the computer program is loaded on the processing unit of the calculator **16**.

[0084] In a variant, the computer program is stored on a cloud accessible through a server.

[0085] Operation of the calculator **16** will now be described with reference to FIG. 2, which diagrammatically illustrates an example of implementation of a method for determining an orientation of the photovoltaic panel **10** to reduce the damage caused by a hail event, and to FIGS. 3 and 4 which illustrate in more detail some notations used when implementing the method.

[0086] The steps of the determination method are, for example, implemented by the calculator **16** in interaction with the computer program product, that is to say are implemented by a computer.

[0087] The determination method comprises a phase **100** for receiving input data relative to characteristics of a hail event.

[0088] The input data comprise at least the mass $m_{\text{sub.hail}}$ of a hailstone and a hailstone velocity vector .

[0089] The hailstone velocity vector  comprises a vertical velocity component $u_{\text{sub.z}}$ and a horizontal velocity component $u_{\text{sub.x}}$. Preferably, the horizontal velocity component is approximated as being the velocity of the wind $u_{\text{sub.x}}$.

[0090] In an example, the vertical velocity component $u_{\text{sub.z}}$ is obtained on the basis of the following formula:

$$[00006] \quad u_z = \sqrt{\frac{8 \times \frac{\rho_{\text{sub.hail}} \times R_{\text{sub.hail}} \times g}{3 \times \frac{\rho_{\text{sub.air}} \times 0.44}}}{}} \quad (1)$$

[0091] Where: [0092] $\rho_{\text{sub.hail}}$ is the density of the hail, [0093] $R_{\text{sub.hail}}$ is the radius of the hail, [0094] g is the acceleration of gravity, and [0095] $\rho_{\text{sub.air}}$ is the density of air.

[0096] In a variant or in complement, the input data are for example obtained through weather forecast or local measurements (onsite measurements carried out in real time or not).

[0097] The determination method comprises a phase **200** for determining an orientation of the photovoltaic panel **10** as a function of the input data, in order to protect elements **12** underneath the photovoltaic panel **10** while preventing damages to occur on the photovoltaic panel **10**.

[0098] The determined orientation corresponds to an intermediate orientation of the photovoltaic panel **10** between a plurality of possible intermediate orientations. Each possible intermediate orientation is such that the angle, called hail impact angle $\theta_{\text{sub.hail,poa}}$, between the normal , to the surface of the photovoltaic panel **10** and the hailstone velocity vector  is strictly comprised between 0 degrees and 90 degrees modulo π .

The orientation of the photovoltaic panel **10** corresponding to a hail impact angle $\theta_{\text{sub.hail,poa}}$ of 0 degrees.

[0099] enables to maximize the protection of elements **12** underneath the photovoltaic panel **10**, without preventing damages to occur on the photovoltaic panel **10** (the kinetic energy E received by the photovoltaic panel **10** is maximized). The orientation of the photovoltaic panel **10** corresponding to a hail impact angle $\theta_{\text{sub.hail,poa}}$ of 90 degrees enables to maximize the prevention of damages on the photovoltaic panel **10**, without protecting elements **12** underneath the photovoltaic panel **10** (the kinetic energy E received by the photovoltaic panel **10** is minimized).

[0100] Preferably, the determination phase **200** comprises the determination of the orientation of the photovoltaic panel **10** so as to fulfill a kinetic energy criterion. The kinetic energy criterion states that the kinetic energy E received by the photovoltaic panel **10** from a hailstone is below a maximum predetermined

kinetic energy $E_{\max}$, enabling to prevent damages to occur on the photovoltaic panel **10** (but without going to an orientation of the photovoltaic panel **10** corresponding to a hail impact angle $\theta_{\text{sub.hail,poa}}$ equal to 90° , because this would damage the crops).

[0101] The maximum predetermined kinetic energy $E_{\max}$ is for example determined experimentally or is obtained from manufacturer data. The maximum predetermined kinetic energy $E_{\max}$ is for example equal to 0.8 Joules. However, this value depends on the type of the photovoltaic panel **10**.

[0102] In particular, the kinetic energy E received by the photovoltaic panel **10** from a hailstone depends on the hail impact angle $\theta_{\text{sub.hail,poa}}$. The greater the hail impact angle $\theta_{\text{sub.hail,poa}}$, the lower the kinetic energy E received by the photovoltaic panel **10**, and vice versa.

[0103] In an example of implementation, the determination phase **200** comprises: [0104] a step of calculating the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$ corresponding to the maximum predetermined kinetic energy $E_{\max}$. For example, the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$ is given by the following formula:

[00007] $u_{\text{sub.hail,poa,max}} = \sqrt{2E_{\max} / m_{\text{hail}}}$ (2) [0105] a step of calculating the hailstone velocity $u_{\text{sub.hail}}$ as a function of the input data. For example, the hailstone velocity $u_{\text{sub.hail}}$ is given by the following formula:

[00008] $u_{\text{sub.hail}} = \sqrt{u_x^2 + u_z^2}$ (3)

where $u_{\text{sub.x}}$ is the horizontal velocity component which is approximated as being the velocity of the wind ($u_{\text{sub.x}} = u_{\text{sub.w}}$) and $u_{\text{sub.z}}$ is the vertical velocity component, both $u_{\text{sub.w}}$ and $u_{\text{sub.z}}$ being given by the input data. [0106] a step of comparing the hailstone velocity $u_{\text{sub.hail}}$ to the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$, and [0107] a step of determining the orientation of the photovoltaic panel **10** as a function of the input data and of the result of the comparison.

[0108] In particular, when the hailstone velocity $u_{\text{sub.hail}}$ is above the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$, the step of determining the orientation of the photovoltaic panel **10** comprises calculating the minimum allowable hailstone impact angle $\theta_{\text{sub.hail,poa,min}}$ on the photovoltaic panel **10**, on the basis of the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$ and the hailstone velocity $u_{\text{sub.hail}}$. The minimum allowable hailstone impact angle $\theta_{\text{sub.hail,poa,min}}$ is for example given by the following formula:

[00009] $\theta_{\text{sub.hail,poa,min}} = \cos^{-1}(\min(1, u_{\text{sub.hail,poa,max}} / u_{\text{sub.hail}}))$ (4)

[0109] The orientation of the photovoltaic panel **10** is then calculated as a function of the minimum allowable hailstone impact angle $\theta_{\text{sub.hail,poa,min}}$ and of the input data.

[0110] Three examples of embodiments corresponding to the calculation of the orientation of the photovoltaic panel **10** as a function of the minimum allowable hailstone impact angle $\theta_{\text{sub.hail,poa,min}}$ and of the input data, will now be described. In addition, FIG. 3 illustrates the notations used in the above calculations and in the calculations hereinafter, POA being the plane of the photovoltaic surface **10A** of the photovoltaic panel **10** and H being a hailstone. FIG. 4 is specific to the third embodiment.

[0111] 1.sup.st embodiment: dual axis tracking at a given $\beta_{\text{sub.poa}}$ (e.g., $\beta_{\text{sub.poa}} = \theta_{\text{sub.hail}}$)

[0112] In this embodiment, the input data also comprise the azimuth $\gamma_{\text{sub.w}}$ of the wind (also called wind direction). Such data is for example obtained through weather forecast or local measurements (onsite measurements carried out in real time or not).

[0113] In the first embodiment, the orientation of the photovoltaic panel **10** is quantified by a tilt angle $\beta_{\text{sub.poa}}$ for the photovoltaic panel **10** and an azimuth angle $\gamma_{\text{sub.poa}}$ for the photovoltaic panel **10** enabling to perform dual axis tracking. The tilt angle $\beta_{\text{sub.poa}}$ is the angle between the normal  to the surface of the photovoltaic panel **10** and a vertical axis. The azimuth angle $\gamma_{\text{sub.poa}}$ is the angle in a horizontal plane between a reference direction (for example, the north) and the projection of the normal  to the surface of the photovoltaic panel **10** in the horizontal plane.

[0114] The step of determining the orientation of the photovoltaic panel **10** comprises fixing one of the two angles and determining the other angle depending on the input data and of the result of the comparison between the hailstone velocity $u_{\text{sub.hail}}$ and the hailstone maximum velocity $u_{\text{sub.hail,poa,max}}$. In this embodiment, the tilt angle $\beta_{\text{sub.poa}}$ of the photovoltaic panel **10** is set at a given value. In an example, the tilt angle $\beta_{\text{sub.poa}}$ is set as being equal to the angle $\theta_{\text{sub.hail}}$ between the hailstone velocity vector  and a vertical axis.

[0115] In this embodiment, when the hailstone velocity $u_{\text{sub.hail}}$ is above the hailstone

[0116] maximum velocity $u_{\text{sub.hail,poa,max}}$, two solutions are determined for the azimuth angle $\gamma_{\text{sub.poa}}$ as a function of the following equation:

$$[00010] \quad \rho_{\text{poa}} = \omega \pm \left(\frac{(\cos(\theta_{\text{hail, poa, min}}) - \cos(\rho_{\text{poa}})\cos(\theta_{\text{hail}}))}{\sin(\rho_{\text{poa}})\sin(\theta_{\text{hail}})} \right) \quad (5)$$

[0117] where: [0118] $\gamma_{\text{sub.poa}}$ is the azimuth angle of the photovoltaic panel **10**, [0119] $\gamma_{\text{sub.w}}$ is the azimuth of the wind, [0120] $\theta_{\text{sub.hail, poa, min}}$ is the minimum allowable hailstone impact angle, [0121] $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector  and a vertical axis, with $\theta_{\text{sub.hail}} = \tan^{-1}(u_{\text{sub.w}}/u_{\text{sub.z}})$, $u_{\text{sub.w}}$ being the horizontal velocity component which is approximated as being the velocity of the wind, and $u_{\text{sub.z}}$ being the vertical velocity component, and [0122] $\beta_{\text{sub.poa}}$ is the tilt angle of the photovoltaic panel **10**.

[0123] Preferably, the azimuth angle $\gamma_{\text{sub.poa}}$ corresponding to the solution the closest from the solar azimuth angle is selected. This enables to minimize the solar angle of incidence, and thus to maximize the photovoltaic power generation.

[0124] When the hailstone velocity $u_{\text{sub.hail}}$ is below the hailstone maximum velocity $U_{\text{sub.hail, poa, max}}$, the azimuth angle $\gamma_{\text{sub.poa}}$ of the photovoltaic panel **10** is set as being equal to the azimuth of the wind $\gamma_{\text{sub.w}}$.

[0125] 2.sup.nd embodiment: dual axis tracking at a given $\gamma_{\text{sub.poa}}$ (e.g., $\gamma_{\text{sub.poa}} = \gamma_{\text{sub.w}}$

[0126] In this embodiment, the input data also comprise the azimuth of the wind $\gamma_{\text{sub.w}}$. Such data is for example obtained through weather forecast or local measurements (onsite measurements carried out in real time or not).

[0127] In the second embodiment, as in the first embodiment, the orientation of the photovoltaic panel **10** is quantified by a tilt angle $\beta_{\text{sub.poa}}$ for the photovoltaic panel **10** and an azimuth angle $\gamma_{\text{sub.poa}}$ for the photovoltaic panel **10**, enabling to perform dual axis tracking.

[0128] The step of determining the orientation of the photovoltaic panel **10** comprises fixing one of the two angles and determining the other angle depending on the input data and of the result of the comparison between the hailstone velocity $u_{\text{sub.hail}}$ and the hailstone maximum velocity $u_{\text{sub.hail, poa, max}}$. In this embodiment, the azimuth angle $\gamma_{\text{sub.poa}}$ of the photovoltaic panel **10** is set at a given value. In an example, the azimuth angle $\gamma_{\text{sub.poa}}$ is set as being equal to the azimuth of the wind $\gamma_{\text{sub.w}}$.

[0129] In this embodiment, when the hailstone velocity $u_{\text{sub.hail}}$ is above the hailstone maximum velocity $u_{\text{sub.hail, poa, max}}$, two solutions are determined for the tilt angle $\beta_{\text{sub.poa}}$ as a function of the following equation:

$$[00011] \quad (6) \quad \cos(\theta_{\text{hail, poa, min}}) = \cos(\rho_{\text{poa}})\cos(\theta_{\text{hail}}) + \sin(\rho_{\text{poa}})\sin(\theta_{\text{hail}})\cos(\gamma_{\text{w}} - \gamma_{\text{poa}})$$

[0130] where: [0131] $\gamma_{\text{sub.poa}}$ is the azimuth angle of the photovoltaic panel **10**, [0132] $\gamma_{\text{sub.w}}$ is the azimuth of the wind, [0133] $\theta_{\text{sub.hail, poa, min}}$ is the minimum allowable hailstone impact angle, $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector  and a vertical axis, and [0134] $\beta_{\text{sub.poa}}$ is the tilt angle of the photovoltaic panel **10**.

[0135] Preferably, the tilt angle $\beta_{\text{sub.poa}}$ corresponding to the solution the closest from the solar zenith angle is selected. This enables to minimize the solar angle of incidence, and thus to maximize the photovoltaic power generation.

[0136] When the hailstone velocity $u_{\text{sub.hail}}$ is below the hailstone maximum velocity $u_{\text{sub.hail, poa, max}}$, the tilt angle $\beta_{\text{sub.poa}}$ of the photovoltaic panel **10** is set as being equal to the hailstone velocity angle $\theta_{\text{sub.hail}}$. The hailstone velocity angle $\theta_{\text{sub.hail}}$ is the angle between the hailstone velocity vector  and the vertical axis. The hailstone velocity angle $\theta_{\text{sub.hail}}$ is obtained from the input data (the hailstone velocity angle $\theta_{\text{sub.hail}}$ is derived from the hailstone velocity vector .

[0137] 3.sup.rd embodiment: single axis tracking with the axis tilted at $\beta_{\text{sub.axis}}$ from horizontal and azimuthally oriented at $\gamma_{\text{sub.axis}}$.

[0138] In the third embodiment, the orientation of the photovoltaic panel **10** is controlled by an angle of rotation ω of a single axis oriented at a tilt angle $\beta_{\text{sub.axis}}$ and at an azimuth angle $\gamma_{\text{sub.axis}}$, enabling to perform single axis tracking.

[0139] When the hailstone velocity $u_{\text{sub.hail}}$ is below the hailstone maximum velocity $u_{\text{sub.hail, poa, max}}$, the hail impact angle $\theta_{\text{sub.hail, poa}}$, between the normal  to the surface of the photovoltaic panel **10** and the hailstone velocity vector , is calculated the following way:

$$[00012] \quad (7)$$

$$\theta_{\text{hail, poa}} = \cos^{-1}(\cos \rho_{\text{prot}}(\cos \alpha_{\text{axis}} \cos \theta_{\text{hail}} + \sin \alpha_{\text{axis}} \sin \theta_{\text{hail}} \cos(\gamma_{\text{w}} - \gamma_{\text{axis}})) + \sin \rho_{\text{prot}} \sin \theta_{\text{hail}} \sin(\gamma_{\text{w}} - \gamma_{\text{axis}})),$$

where $\rho_{\text{prot}} = \tan^{-1}(X) + \beta_{\text{sub.axis}} = \tan^{-1}\left(\frac{\sin \theta_{\text{hail}} \cos(\gamma_{\text{w}} - \gamma_{\text{axis}})}{\cos \alpha_{\text{axis}} \cos \theta_{\text{hail}} + \sin \alpha_{\text{axis}} \sin \theta_{\text{hail}} \cos(\gamma_{\text{w}} - \gamma_{\text{axis}})}\right) + \beta_{\text{sub.axis}}$, where

$$= \begin{cases} 0^\circ, \text{if } X = 0, \text{or if } X > 0 \text{ and } (\omega_w - \alpha_{axis}) > 0, \\ \text{or if } X < 0 \text{ and } (\omega_w - \alpha_{axis}) < 0 \\ +180^\circ, \text{if } X < 0 \text{ and } (\omega_w - \alpha_{axis}) > 0 \\ -180^\circ, \text{if } X > 0 \text{ and } (\omega_w - \alpha_{axis}) < 0 \end{cases}$$

[0140] where: [0141] $\omega_{sub.prot}$ is the protection angle of rotation of the single axis, [0142] $\beta_{sub.axis}$ is the tilt angle of the single axis, [0143] $\gamma_{sub.axis}$ is the azimuth angle of the single axis, [0144] $\gamma_{sub.w}$ is the azimuth of the wind, and [0145] $\theta_{sub.hail}$ is the angle between the hailstone velocity vector

 and a vertical axis.

[0146] When the hailstone velocity $u_{sub.hail}$ is below the hailstone maximum velocity $u_{sub.hail,max}$, and consequently, when the hail impact angle $\theta_{sub.hail,poa}$, between the normal  to the surface of the photovoltaic panel **10** and the hailstone velocity vector , is always greater than or equal to the minimum allowable hailstone impact angle $\theta_{sub.hail,poa,min}$, the rotation angle of the single axis is set as being equal to the value $\omega_{sub.prot}$.

[0147] When the hail impact angle $\theta_{sub.hail,poa}$, between the normal  to the surface of the photovoltaic panel **10**, which is rotated at $\omega_{sub.prot}$, and the hailstone velocity vector , below the minimum allowable hailstone impact angle $\theta_{sub.hail,poa,min}$, two solutions are determined for the rotation angle ω of the single axis as a function of the following equation:

[00013] (8)

$$\theta_{sub.hail,poa} = \cos^{-1}(\cos(\omega_{axis} \cos \theta_{sub.hail} + \sin \omega_{axis} \sin \theta_{sub.hail} \cos(\omega_w - \alpha_{axis})) + \sin \omega_{axis} \sin \theta_{sub.hail} \sin(\omega_w - \alpha_{axis}))$$

[0148] where: [0149] ω is the angle of rotation of the single axis.

[0150] Preferably, out of the two solutions the one that minimizes the solar angle of incidence is selected. This enables to maximize the photovoltaic power generation.

[0151] Hence, each of these three embodiments enable to obtain an orientation for the photovoltaic panel **10** with different piloting modes, so as to prevent hail damage.

[0152] In operation, the determination method also comprises a phase **300** for sending a command to the actioner **14** of the photovoltaic panel **10** enabling to adapt the orientation of the photovoltaic panel **10** in conformity with the determined orientation. Such an actioner **14** then modifies the orientation of the photovoltaic panel.

[0153] Such a method could be applied to a whole array of photovoltaic panels.

[0154] Hence, the determination method enables to determine an orientation of the photovoltaic panel **10** enabling to reduce the damages caused by a hail event on both the photovoltaic panel **10** and crops. In particular, a balance is found between the prevention of the damages of the photovoltaic panel **10** and the prevention of the damages of the crops or any other elements underneath the photovoltaic panel **10**.

[0155] The person skilled in the art will understand that the embodiments and variants described above can be combined to form new embodiments provided that they are technically compatible.

Claims

1. A method for determining an orientation of a photovoltaic panel to reduce the damage caused by a hail event, the method being computer-implemented and comprising: receiving input data relative to characteristics of a hail event, the input data comprising at least the mass of a hailstone and a hailstone velocity vector, and determining an orientation of the photovoltaic panel as a function of the input data, in order to protect elements underneath the photovoltaic panel while preventing damages to occur on the photovoltaic panel, the determined orientation corresponding to an intermediate orientation of the photovoltaic panel between a plurality of possible intermediate orientations, each possible intermediate orientation being such that the angle, called hail impact angle, between the normal to the surface of the photovoltaic panel and the hailstone velocity vector, is strictly comprised between 0 degrees and 90 degrees modulo π , the orientation of the photovoltaic panel corresponding to a hail impact angle of 0 degrees enabling to maximize the protection of elements underneath the photovoltaic panel without preventing damages to occur on the photovoltaic panel, the orientation of the photovoltaic panel corresponding to a hail impact angle of 90 degrees enabling to maximize the prevention of damages on the photovoltaic panel without protecting elements underneath the photovoltaic panel.

2. The method according to claim 1, wherein the method comprises a phase of sending a command to an actioner of the photovoltaic panel enabling to adapt the orientation of the photovoltaic panel in conformity with the determined orientation.

3. The method according to claim 1, wherein the determination phase comprises the determination of the orientation of the photovoltaic panel so as to fulfill a kinetic energy criterion, the kinetic energy criterion stating that the kinetic energy received by the photovoltaic panel from a hailstone is below a maximum predetermined kinetic energy, enabling to prevent damages to occur on the photovoltaic panel.

4. The method according to claim 3, wherein the determination phase comprises the steps of: calculating the hailstone maximum velocity corresponding to the maximum predetermined kinetic energy, calculating the hailstone velocity as a function of the input data, comparing the hailstone velocity to the hailstone maximum velocity, and determining the orientation of the photovoltaic panel as a function of the input data and of the result of the comparison.

5. The method according to claim 4, wherein when the hailstone velocity is above the hailstone maximum velocity, the step of determining the orientation of the photovoltaic panel comprises calculating the minimum allowable hailstone impact angle on the basis of the hailstone maximum velocity and the hailstone velocity, and calculating the orientation of the photovoltaic panel as a function of the minimum allowable hailstone impact angle and of the input data.

6. The method according to claim 5, wherein the orientation of the photovoltaic panel is quantified by a tilt angle for the photovoltaic panel and an azimuth angle for the photovoltaic panel enabling to perform dual axis tracking, the step of determining the orientation of the photovoltaic panel comprising fixing one of the two angles and determining the other angle depending on the input data and of the result of the comparison.

7. The method according to claim 6, wherein the input data also comprise the azimuth of the wind, the step of determining the orientation of the photovoltaic panel comprising setting the tilt angle of the photovoltaic panel at a given value, and when the hailstone velocity is above the hailstone maximum velocity, determining two solutions for the azimuth angle as a function of the following equation:

$$\text{poa} = \text{w} \pm \left(\frac{(\cos(\text{hail, poa, min}) - \cos(\text{poa})\cos(\text{hail}))}{\sin(\text{poa})\sin(\text{hail})} \right)$$
 where: $\gamma.\text{sub.poa}$ is the azimuth angle of the photovoltaic panel, $\gamma.\text{sub.w}$ is the azimuth of the wind, $\theta.\text{sub.hail, poa, min}$ is the minimum allowable hailstone impact angle, $\theta.\text{sub.hail}$ is the angle between the hailstone velocity vector and a vertical axis, and $\beta.\text{sub.poa}$ is the tilt angle of the photovoltaic panel, when the hailstone velocity is below the hailstone maximum velocity, setting the azimuth angle of the photovoltaic panel as being equal to the azimuth of the wind.

8. The method according to claim 7, wherein the tilt angle of the photovoltaic panel is set as being equal to the hailstone velocity angle, the hailstone velocity angle being the angle between the hailstone velocity vector and a vertical axis.

9. The method according to claim 6, wherein the input data also comprise the azimuth of the wind, the step of determining the orientation of the photovoltaic panel comprising setting the azimuth angle of the photovoltaic panel at a given value, and when the hailstone velocity is above the hailstone maximum velocity, determining two solutions for the tilt angle as a function of the following equation:

$$\cos(\text{hail, poa, min}) = \cos(\text{poa})\cos(\text{hail}) + \sin(\text{poa})\sin(\text{hail})\cos(\text{w} - \text{poa})$$
 where: $\gamma.\text{sub.poa}$ is the azimuth angle of the photovoltaic panel, $\gamma.\text{sub.w}$ is the azimuth of the wind, $\theta.\text{sub.hail, poa, min}$ is the minimum allowable hailstone impact angle, $\theta.\text{sub.hail}$ is the angle between the hailstone velocity vector and a vertical axis, and $\beta.\text{sub.poa}$ is the tilt angle of the photovoltaic panel, when the hailstone velocity is below the hailstone maximum velocity, setting the tilt angle of the photovoltaic panel as being equal to the hailstone velocity angle, the hailstone velocity angle being the angle between the hailstone velocity vector and a vertical axis.

10. The method according to claim 9, wherein the azimuth angle of the photovoltaic panel is set as being equal to the azimuth of the wind.

11. The method according to claim 5, wherein the orientation of the photovoltaic panel is quantified by an angle of rotation of a single axis oriented at a tilt angle and at an azimuth angle, enabling to perform single axis tracking, the step of determining the orientation of the photovoltaic panel comprising: when the hailstone velocity is below the hailstone maximum velocity, the hail impact angle, between the normal, to the surface of the photovoltaic panel and the hailstone velocity vector is calculated the following way:

$$\text{hail, poa} = \cos^{-1}(\cos \text{prot} (\cos \text{axis} \cos \text{hail} + \sin \text{axis} \sin \text{hail} \cos(\text{w} - \text{axis}))) + \sin \text{prot} \sin \text{hail} \sin(\text{w} - \text{axis})),$$

where $\text{prot} = \tan^{-1}(X) + \dots = \tan^{-1}\left(\frac{\sin \text{hail} \cos(\text{w} - \text{axis})}{\cos \text{axis} \cos \text{hail} + \sin \text{axis} \sin \text{hail} \cos(\text{w} - \text{axis})}\right) + \dots$, where

$$\begin{aligned}
& 0^\circ, \text{ if } X = 0, \text{ or if } X > 0 \text{ and } (\omega_{\text{sub}} - \beta_{\text{sub}}) > 0, \\
& \text{ or if } X < 0 \text{ and } (\omega_{\text{sub}} - \beta_{\text{sub}}) < 0 \\
= \{ & +180^\circ, \text{ if } X < 0 \text{ and } (\omega_{\text{sub}} - \beta_{\text{sub}}) > 0 \\
& -180^\circ, \text{ if } X > 0 \text{ and } (\omega_{\text{sub}} - \beta_{\text{sub}}) < 0
\end{aligned}$$

where: $\omega_{\text{sub}}.$ prot is the protection angle of rotation of

the single axis,. $\beta_{\text{sub}}.$ axis is the tilt angle of the single axis, $\gamma_{\text{sub}}.$ axis is the azimuth angle of the single axis, $\gamma_{\text{sub}}.$ w is the azimuth of the wind, $\theta_{\text{sub}}.$ hail,poa,min is the minimum allowable hailstone impact angle, and $\theta_{\text{sub}}.$ hail is the angle between the hailstone velocity vector and a vertical axis, when the hailstone velocity is below the hailstone maximum velocity, and consequently, when the hail impact angle, between the normal, to the surface of the photovoltaic panel and the hailstone velocity vector, is always greater than or equal to the minimum allowable hailstone impact angle, the rotation angle of the single axis is set as being equal to the value. when the hail impact angle, between the normal to the surface of the photovoltaic panel, which is rotated at $\omega_{\text{sub}}.$ prot, and the hailstone velocity vector, is below the minimum allowable hailstone impact angle, two solutions are determined for the rotation angle ω of the single axis as a function of the following equation:

$\theta_{\text{sub}}.$ hail,poa,min = $\cos^{-1}(\cos \omega (\cos \beta_{\text{sub}}.$ axis $\cos \theta_{\text{sub}}.$ hail + $\sin \beta_{\text{sub}}.$ axis $\sin \theta_{\text{sub}}.$ hail $\cos(\gamma_{\text{sub}}.$ w - $\gamma_{\text{sub}}.$ axis)) + $\sin \omega \sin \theta_{\text{sub}}.$ hail $\sin(\gamma_{\text{sub}}.$ w - $\gamma_{\text{sub}}.$ axis)) where: ω is the angle of rotation of the single axis.

12. The method according to claim 1, wherein the hailstone velocity vector comprises a vertical velocity component and a horizontal velocity component, the horizontal velocity component being approximated as being the velocity of the wind.

13. The method according to claim 12, wherein the vertical velocity component is obtained on the basis of the following formula: $u_z = \sqrt{\frac{8 \times \rho_{\text{hail}} \times R_{\text{hail}} \times g}{3 \times \rho_{\text{air}} \times 0.44}}$ Where: $\rho_{\text{sub}}.$ hail is the density of the hail, $R_{\text{sub}}.$ hail is the radius of the hail, g is the acceleration of gravity, and $\rho_{\text{sub}}.$ air is the density of air.

14. A readable information carrier on which a computer program product comprising a readable information carrier having stored thereon a computer program comprising program instructions, the computer program being loadable onto a data processing unit and causing a method according to claim 1 to be carried out when the computer program is carried out on the data processing unit.
